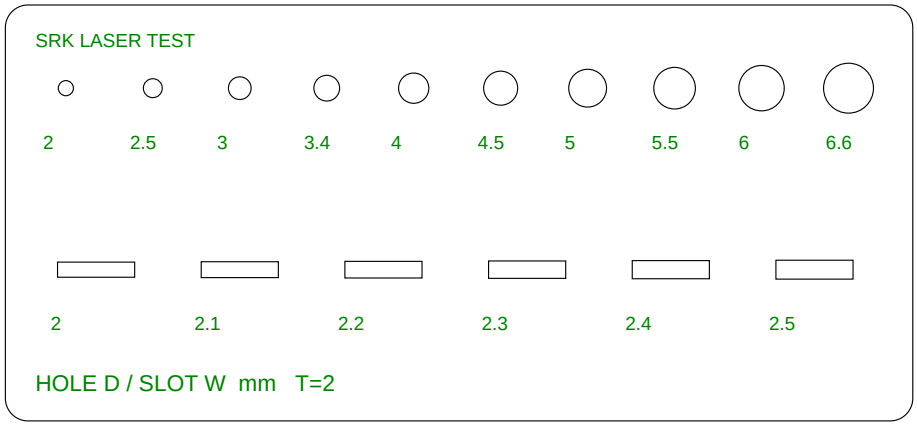
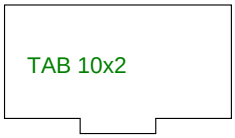
01 Test coupon: hole sizes, slot widths, tab fit

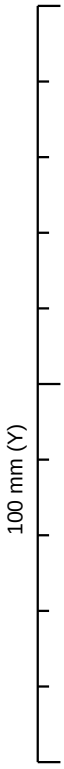
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

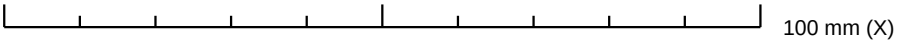


01-test-coupon



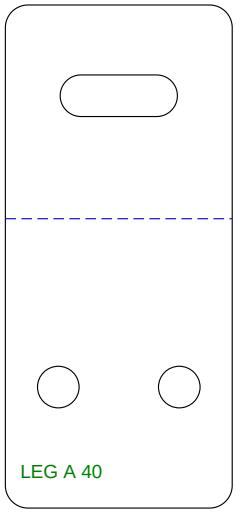
01-test-tab



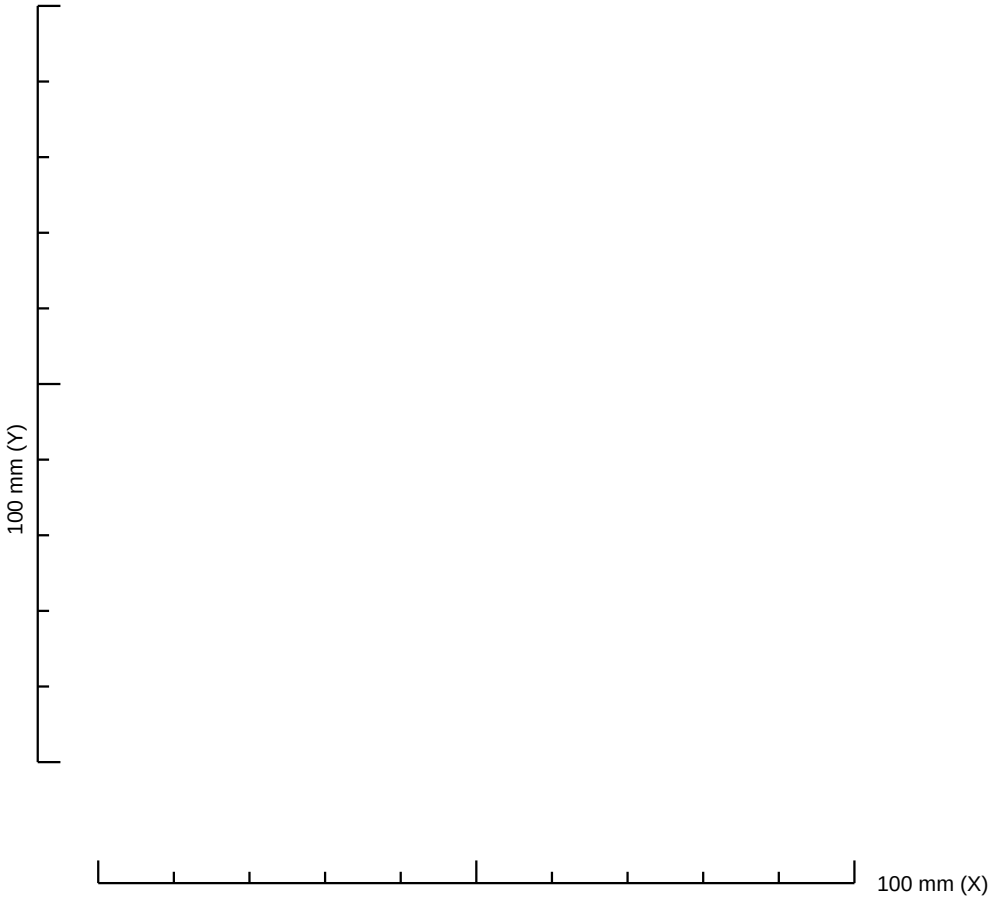


02 L-bracket with bend line (flat pattern)

PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

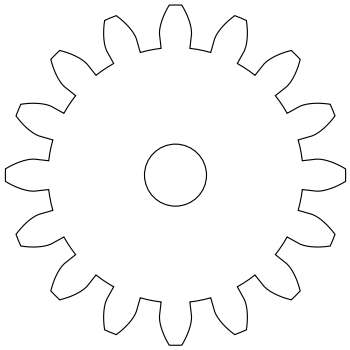


02-l-bracket

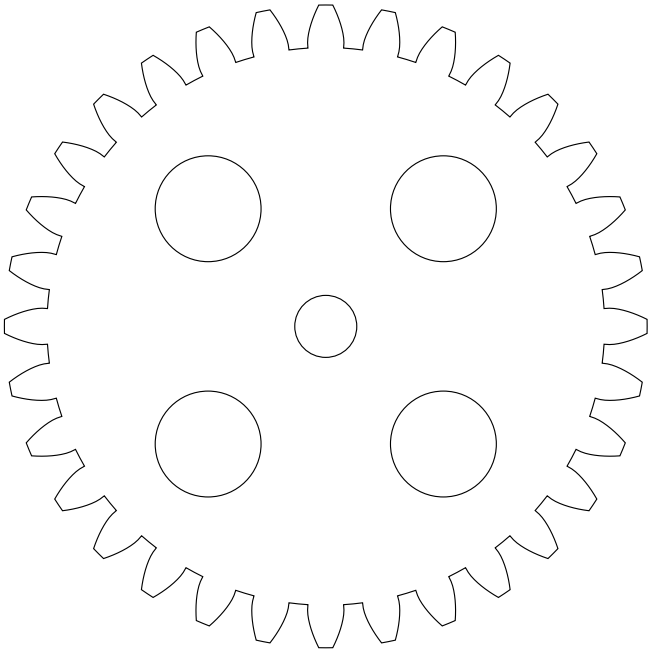


03 Spur gear pair m2.5, 16T / 32T, and axle plate

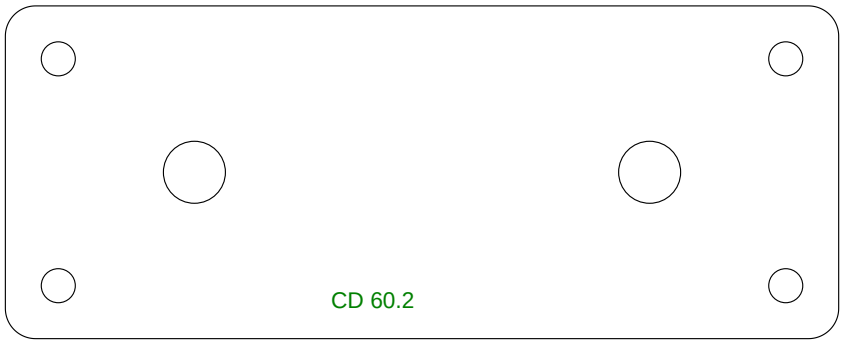
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.



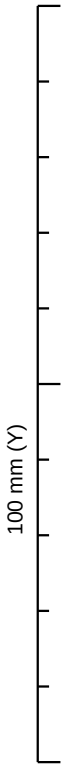
03-gear-16t

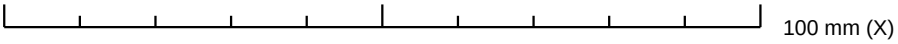


03-gear-32t



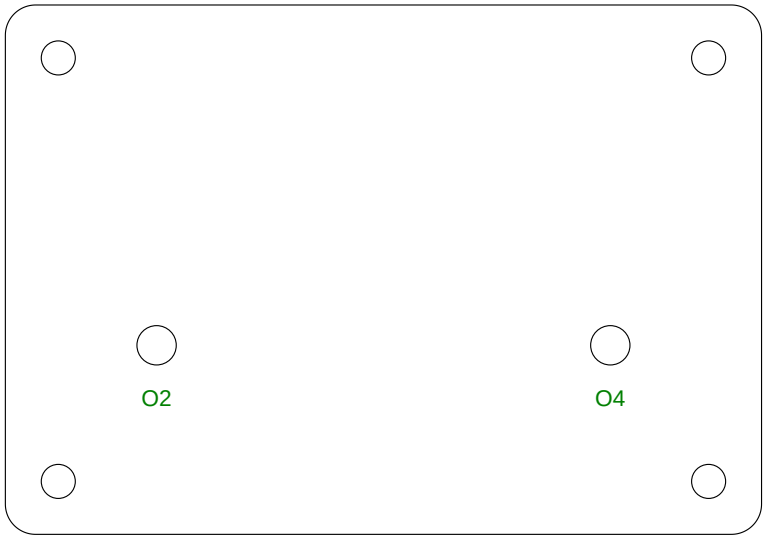
03-gear-plate





04 Four-bar linkage: base and three links

PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.



04-four-bar-base



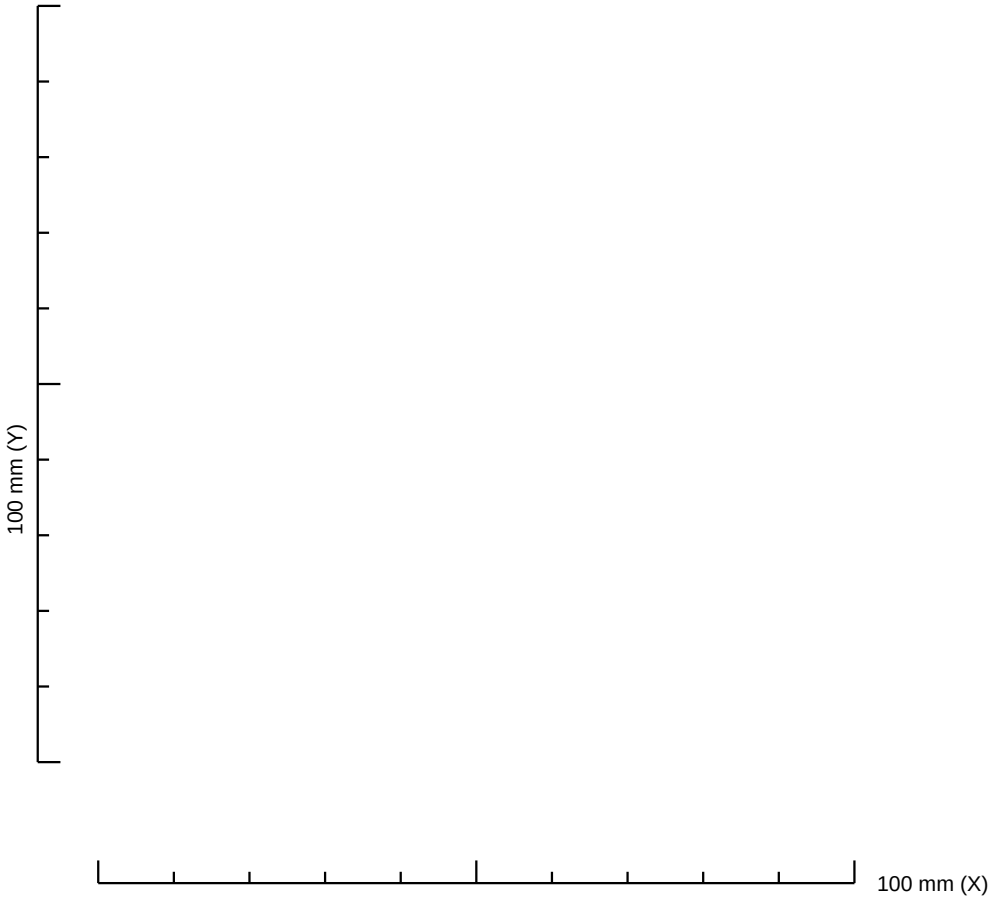
04-four-bar-crank



04-four-bar-coupler

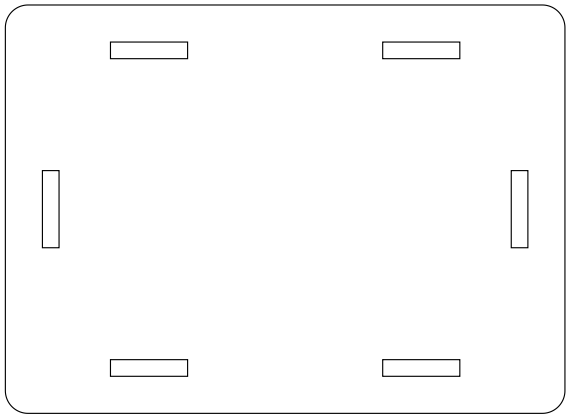


04-four-bar-rocker

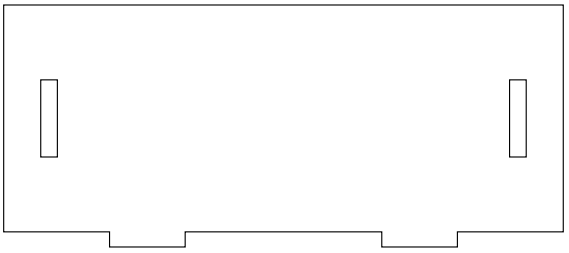


05 Tab-and-slot box, 60 x 40 x 30 inside

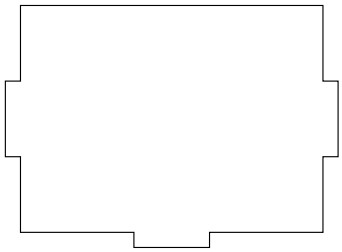
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.



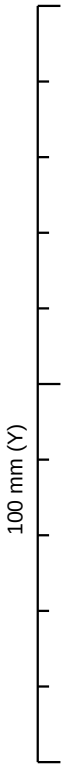
05-box-bottom

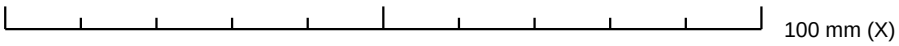


05-box-front-back x2



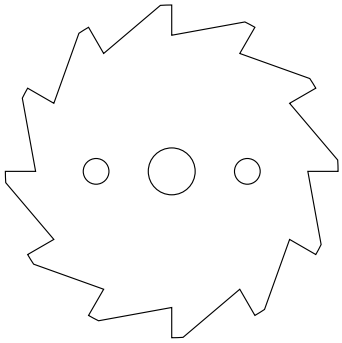
05-box-side x2



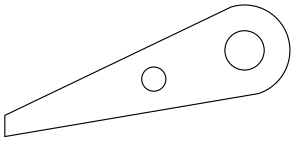


06 Ratchet wheel, pawl and base plate

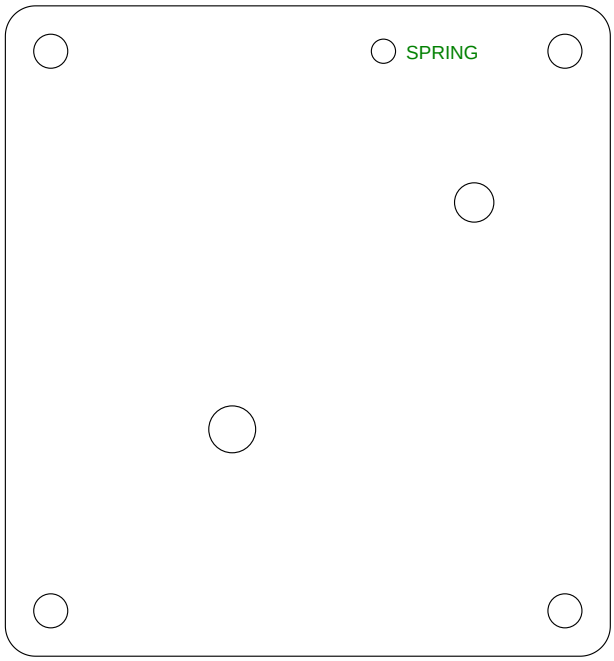
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.



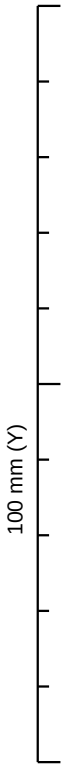
06-ratchet-wheel

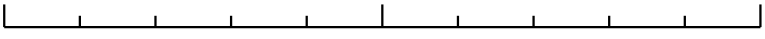


06-ratchet-pawl



06-ratchet-base



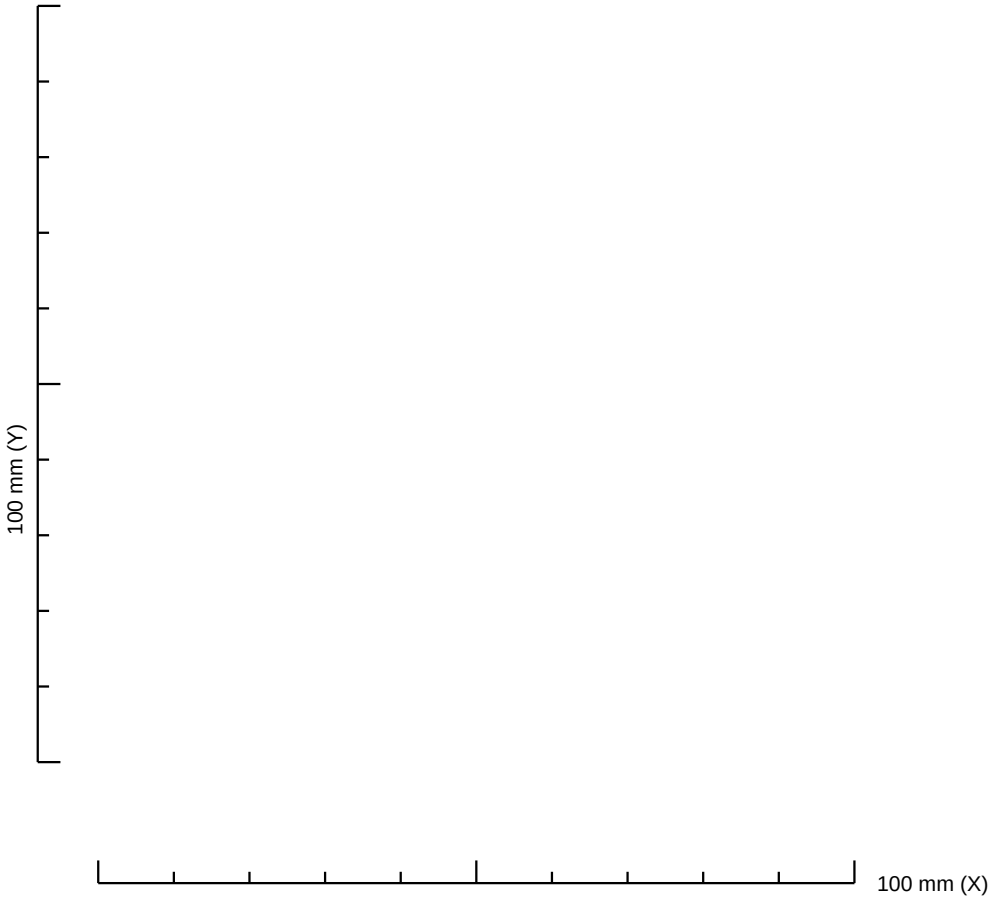


07 Name plate with engraving

PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.



07-nameplate

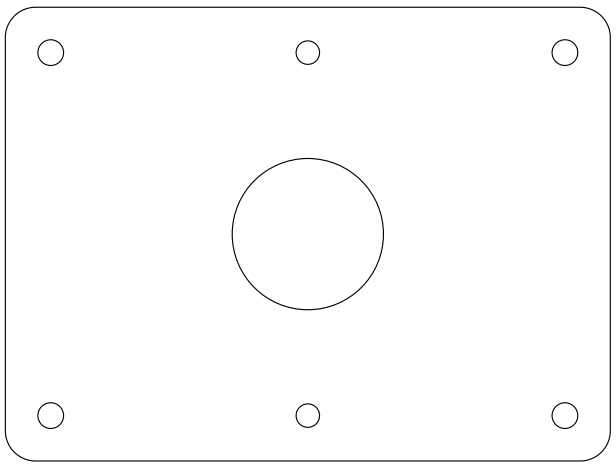


08 Two-plate sandwich (the basic pattern)

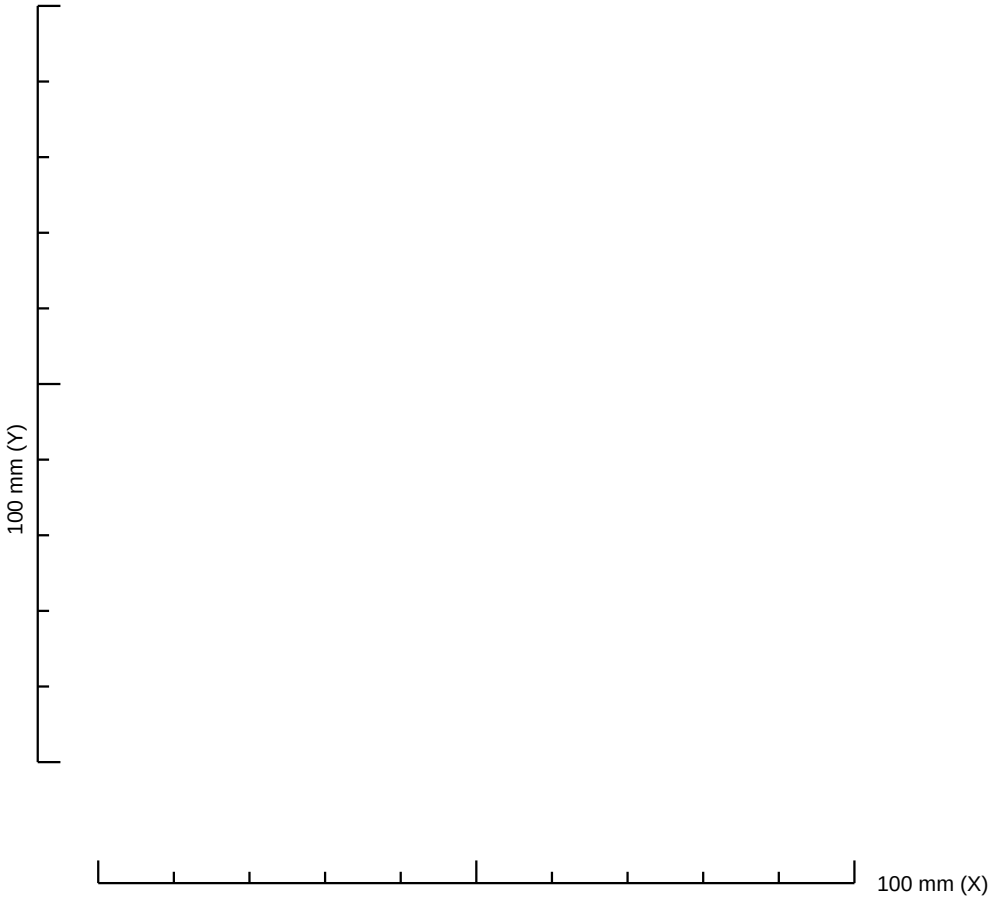
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

- Gap 30 mm. 4x M3 threaded rod 42 mm, 16x M3 nut, 16x M3 washer
- 2x 3 mm smooth rod 34 mm (locating pins, snug fit)



08-sandwich-plate x2

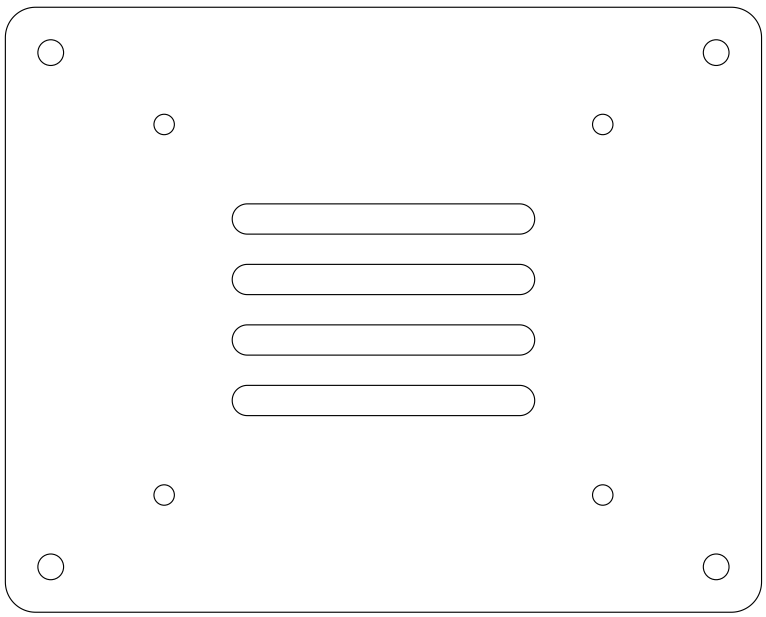


09 Electronics stack (Raspberry Pi 4 pattern)

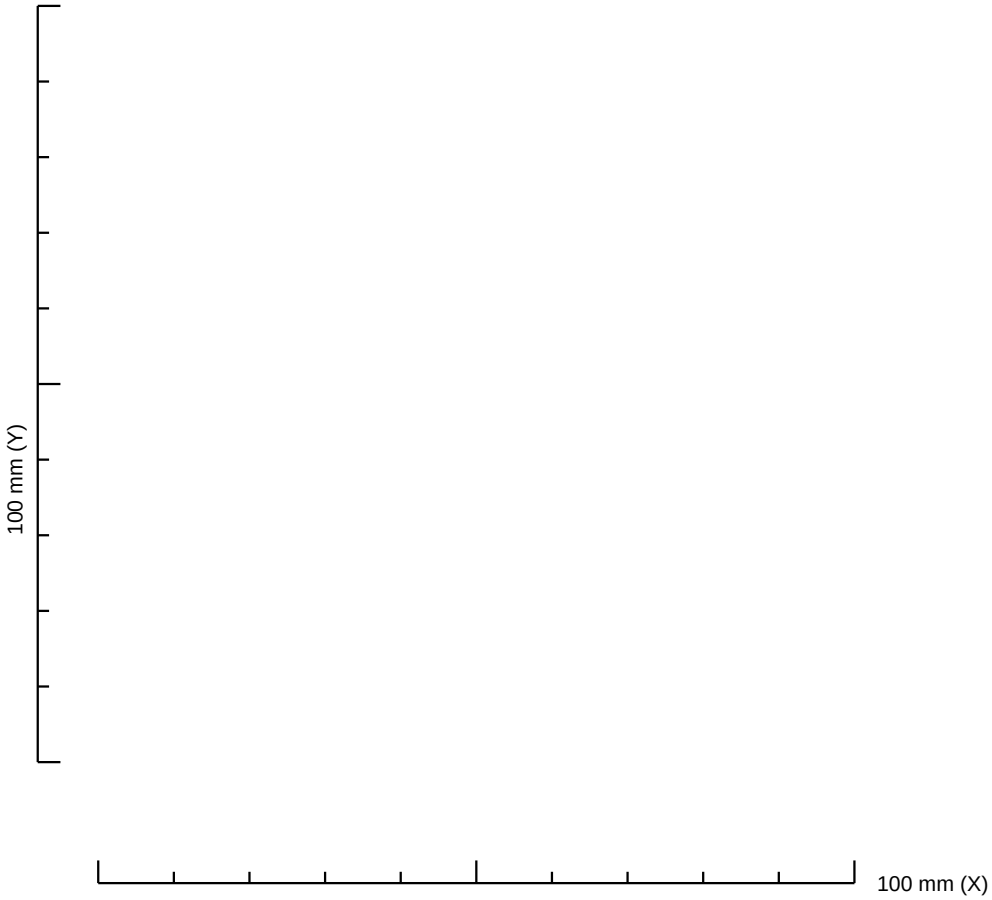
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

- Gap 25 mm. 4x M3 threaded rod 37 mm, 16x M3 nut, 16x M3 washer
- 4x M2.5 x 11 standoffs + 8x M2.5 x 6 screws for the board



09-electronics-plate x2

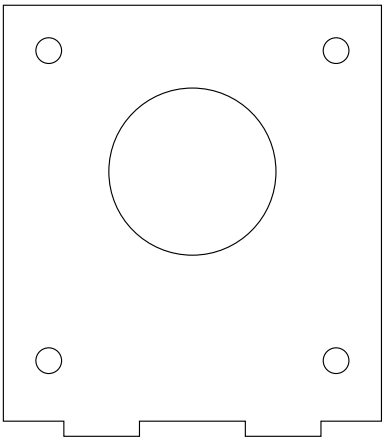


10 Bearing block: two 608 bearings, 8 mm shaft

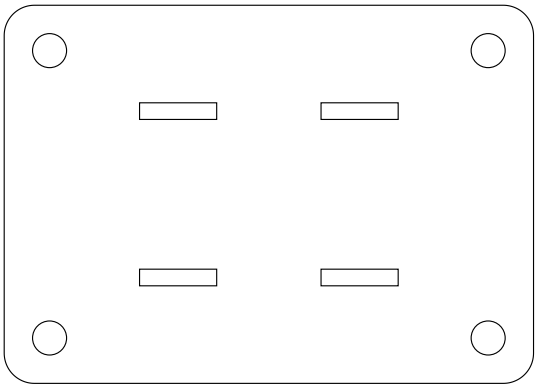
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

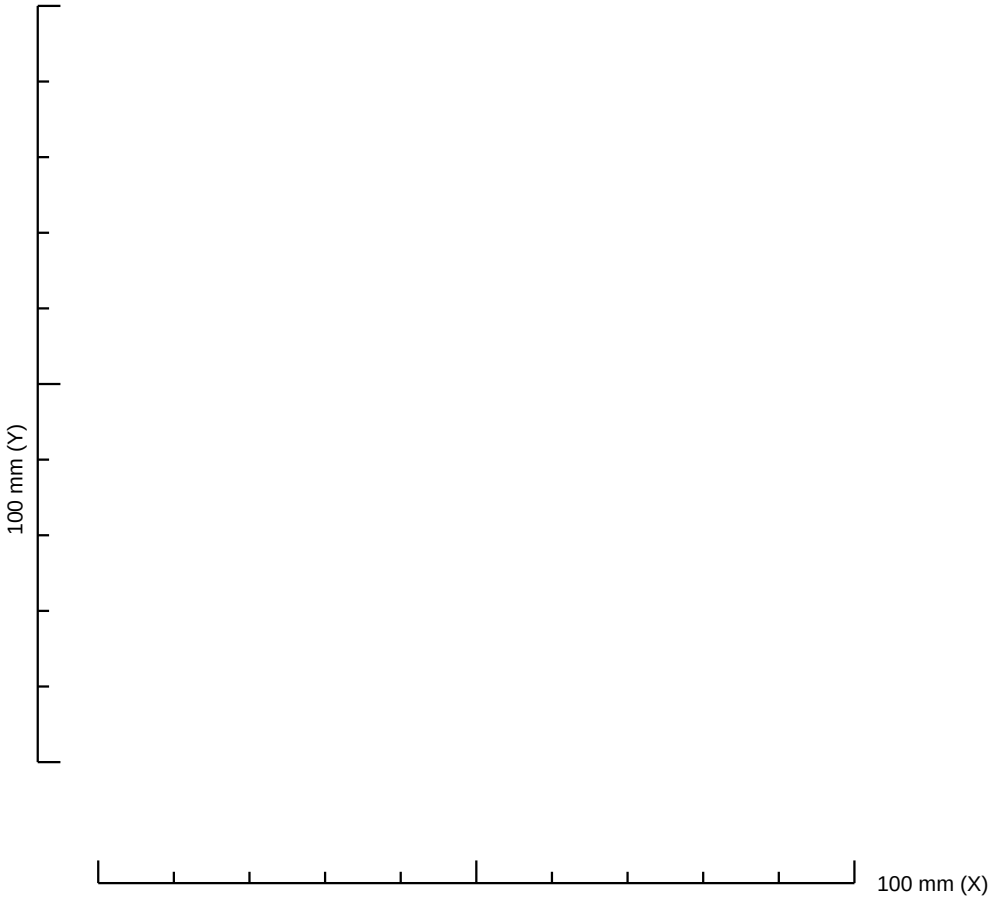
- Gap 20 mm. 4x M3 threaded rod 32 mm, 16x M3 nut, 16x M3 washer
- 2x 608 bearing (F608 flanged is easier), 8 mm shaft, retaining compound



10-bearing-upright x2



10-bearing-base

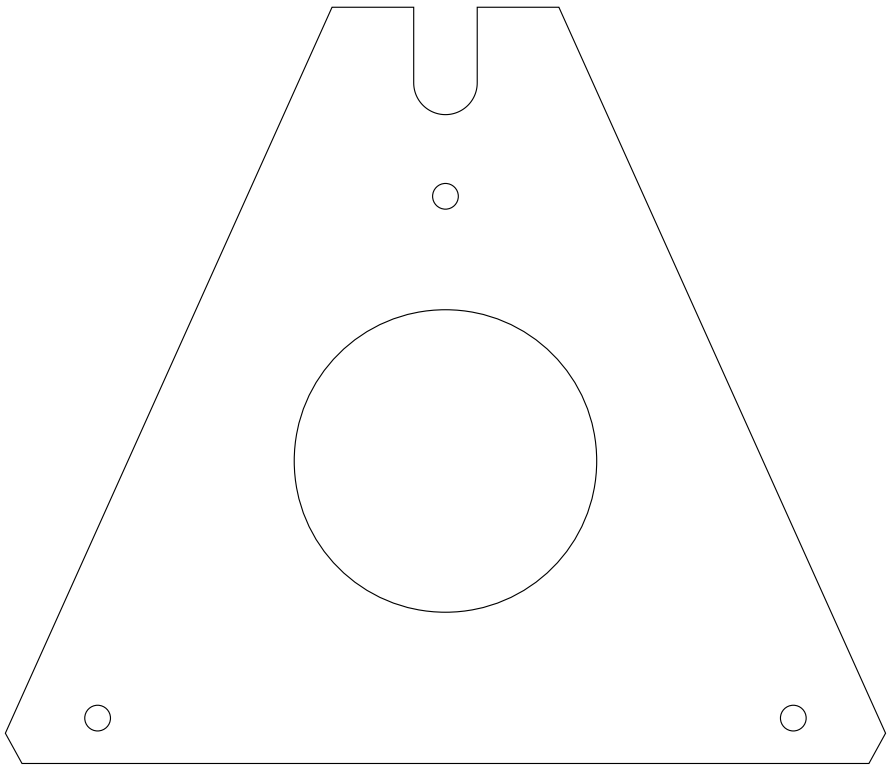


11 Spool holder with 8 mm axle

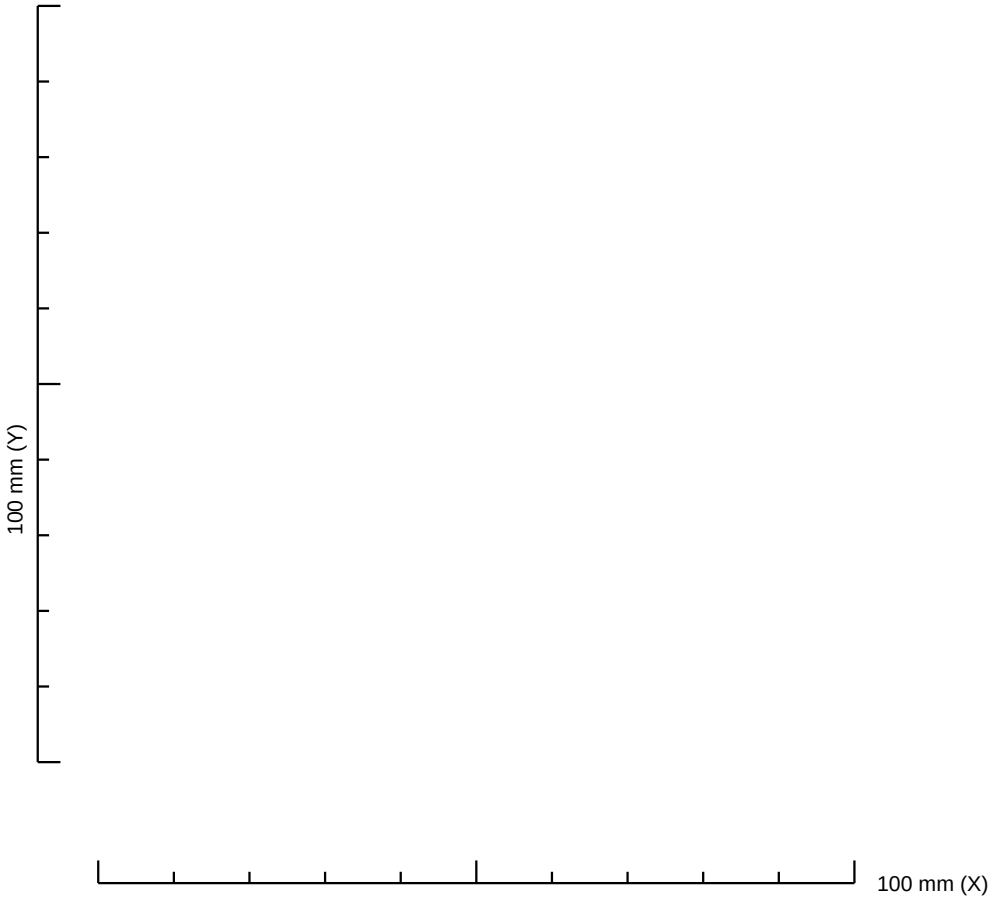
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

- Gap 80 mm. 3x M3 threaded rod 92 mm, 12x M3 nut, 12x M3 washer
- 1x 8 mm smooth rod 100 mm (spool axle)



11-spool-side x2

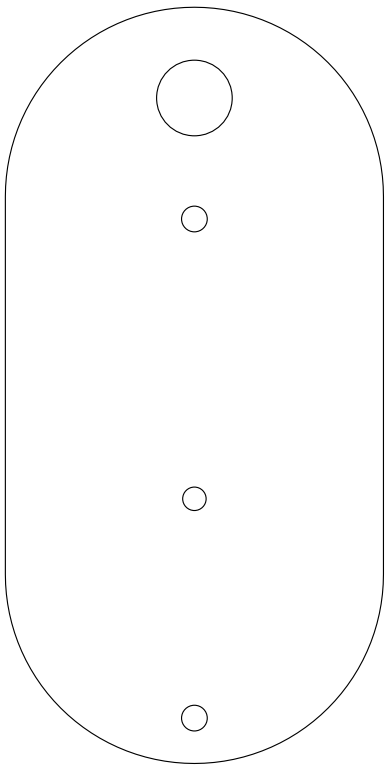


12 Rope pulley block

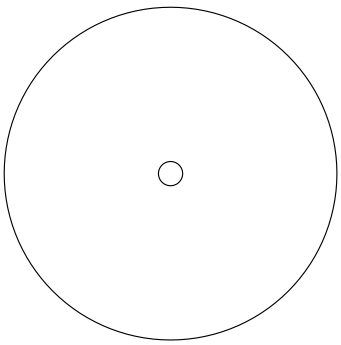
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

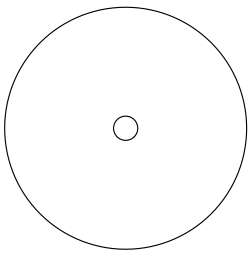
- Gap 10 mm. 2x M3 threaded rod 22 mm, 8x M3 nut, 8x M3 washer
- 1x 3 mm smooth rod 14 mm (axle, glue or e-clips); rope up to 3.5 mm; shackle



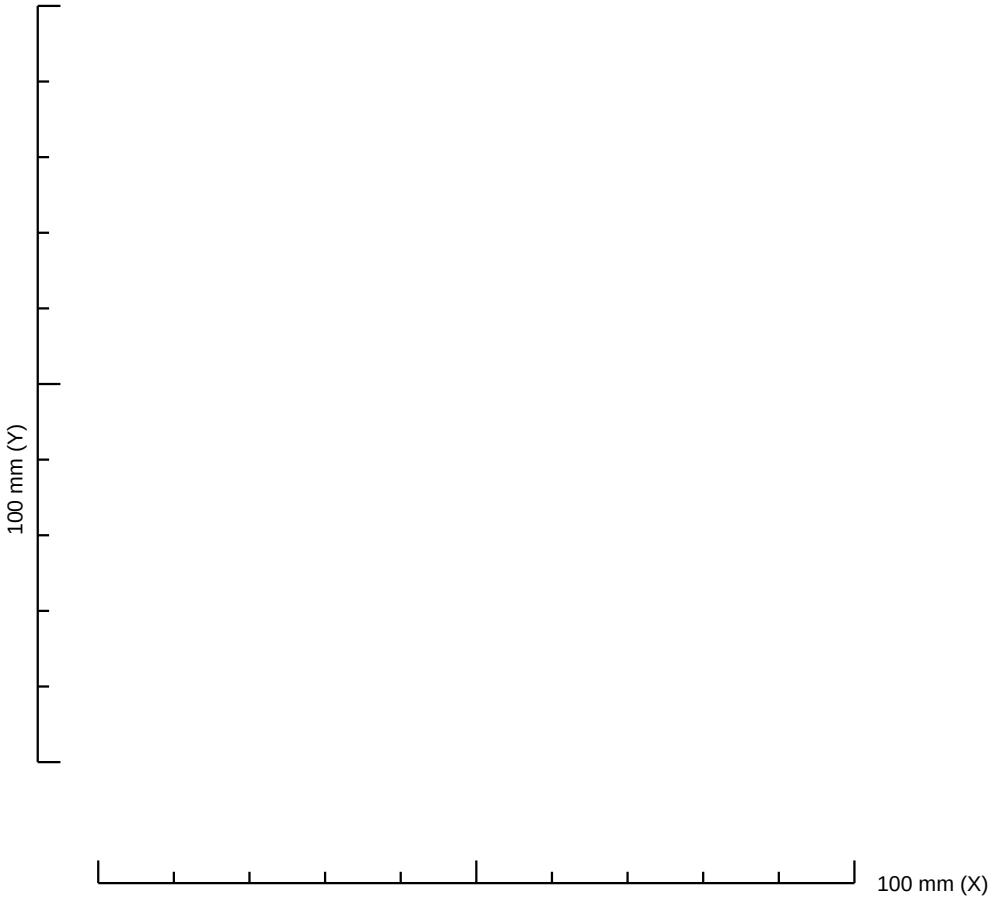
12-pulley-cheek x2



12-sheave-flange x2



12-sheave-core x2

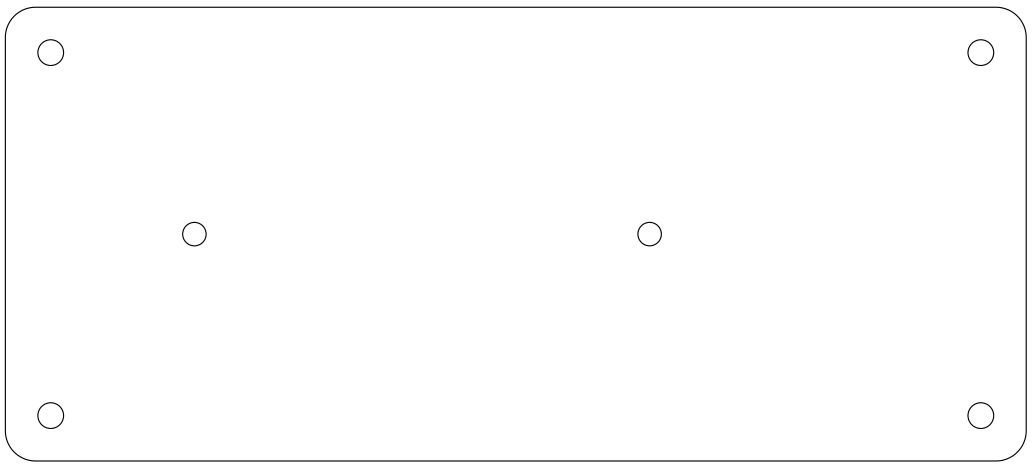


13 Gear train between plates (16T / 32T)

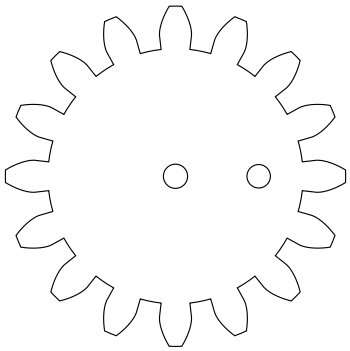
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

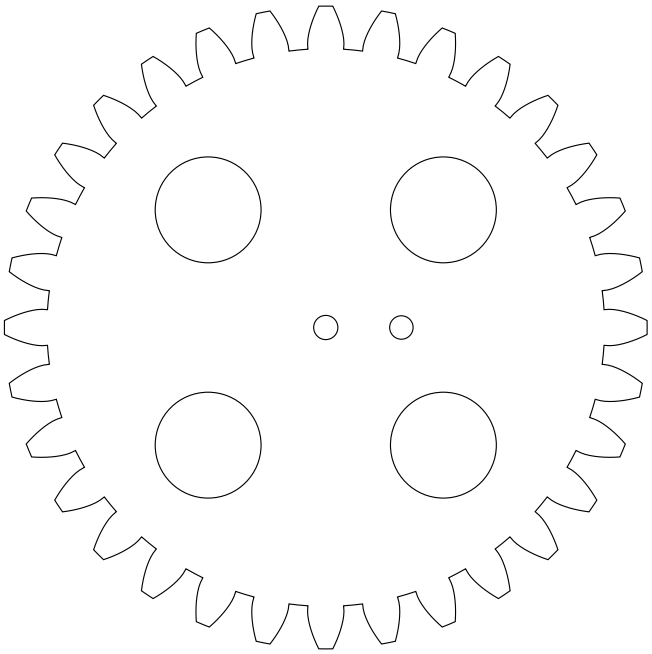
- Gap 10 mm. 4x M3 threaded rod 22 mm, 16x M3 nut, 16x M3 washer
- 2x 3 mm smooth rod 14 mm (axles), 2x 3 mm rod 6 mm (lamination pins), 4x washer



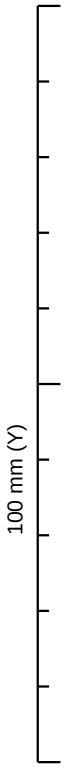
13-gear-plate x2

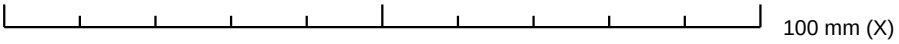


13-gear-16t x3



13-gear-32t x3



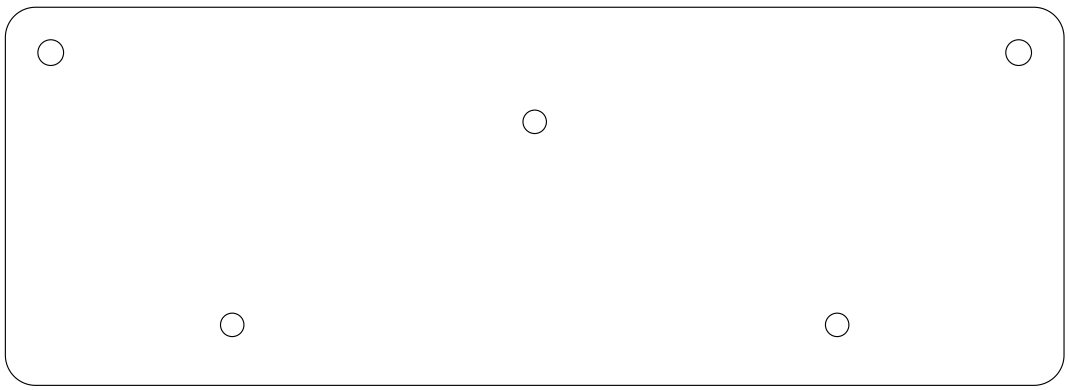


14 Rack and pinion, rack sliding on rods

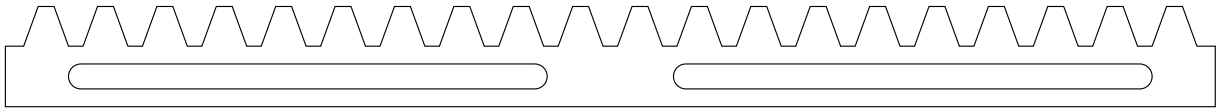
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

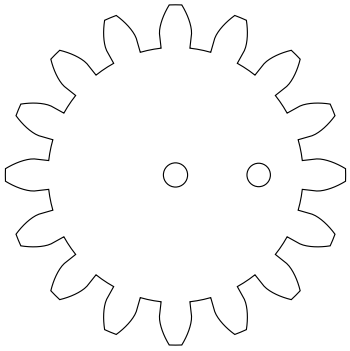
- Gap 10 mm. 2x M3 threaded rod 22 mm, 8x M3 nut, 8x M3 washer
- 3x 3 mm smooth rod 14 mm (2 rack guides + pinion axle), 1x 3 mm rod 6 mm (pin)



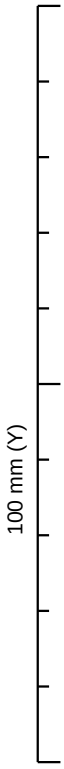
14-rack-plate x2

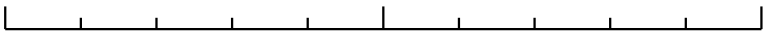


14-rack x3



14-pinion-16t x3



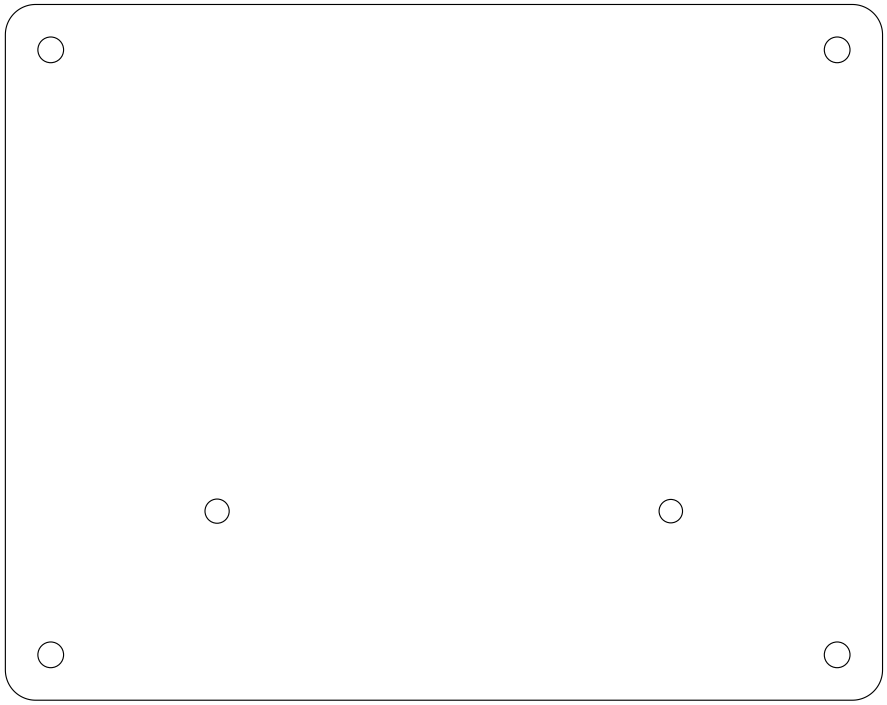


15 Four-bar linkage between plates

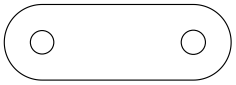
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

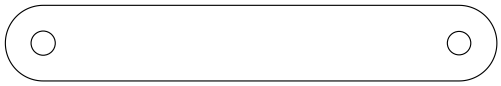
- Gap 8 mm. 4x M3 threaded rod 20 mm, 16x M3 nut, 16x M3 washer
- 3 mm smooth rod: 1x 9 mm crank stub (handle + bottom plate + crank), 1x 5 mm
- rocker stub (bottom plate + rocker), 2x 5 mm coupler pins, 1x 15 mm handle grip



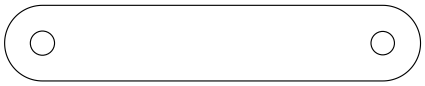
15-four-bar-plate x2



15-four-bar-crank

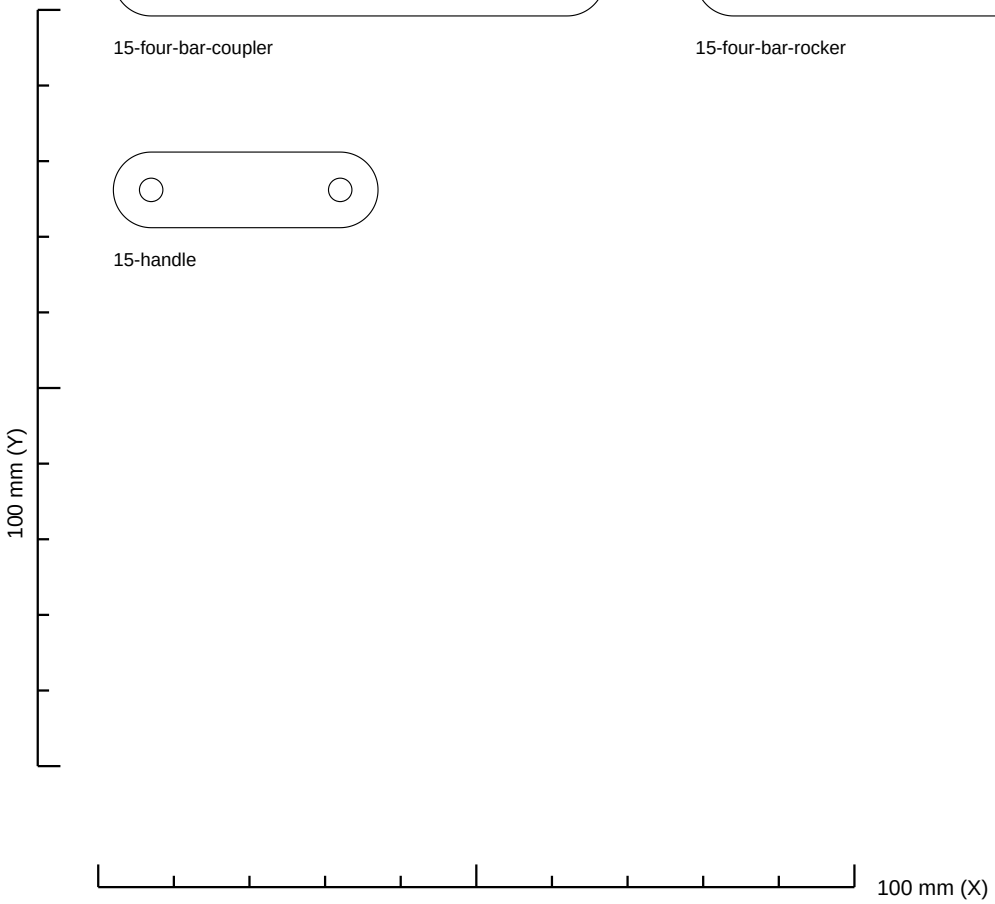


15-four-bar-coupler



15-four-bar-rocker

15-handle

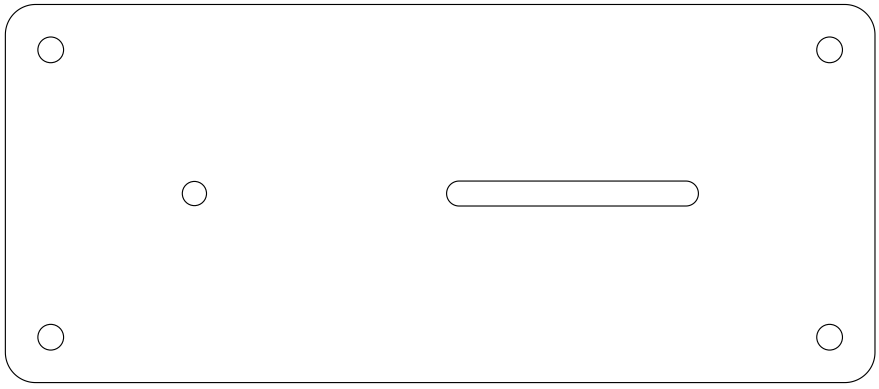


16 Slider-crank with slot-guided slider

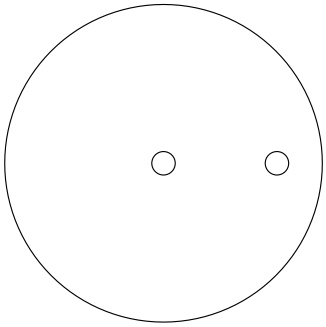
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

- Gap 8 mm. 4x M3 threaded rod 20 mm, 16x M3 nut, 16x M3 washer
- 3 mm smooth rod: 1x 9 mm crank stub (handle + bottom plate + disc), 1x 5 mm crank
- pin, 1x 16 mm slider pin (both slots, e-clips), 1x 15 mm handle grip



16-slider-crank-plate x2



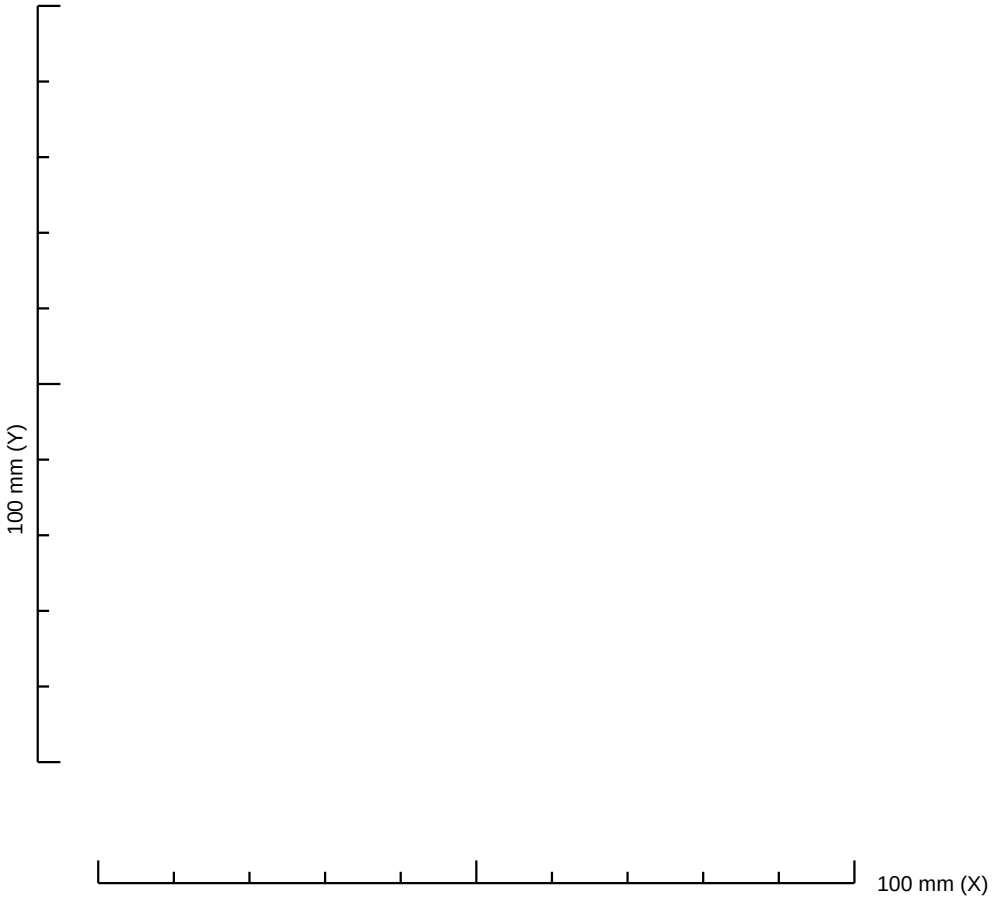
16-crank-disc



16-con-rod



16-handle

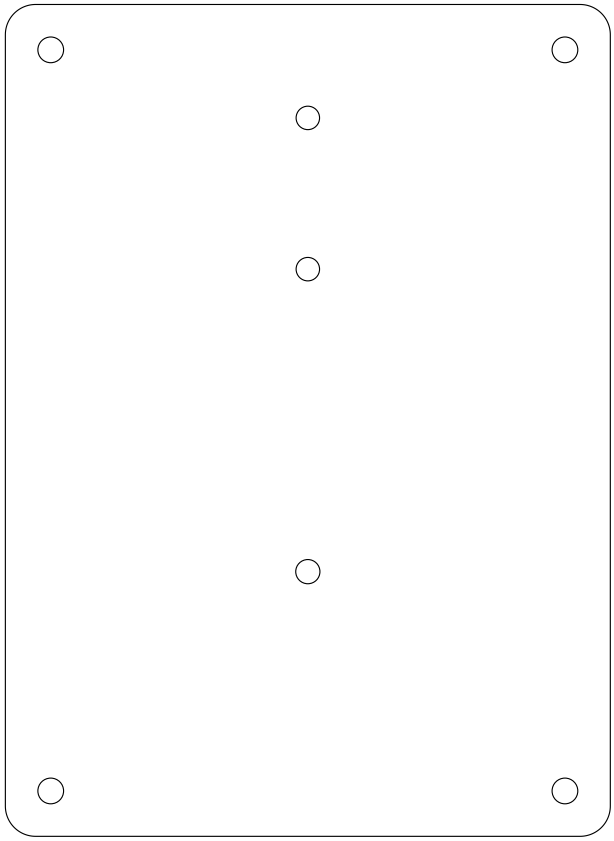


17 Cam and rod-guided follower

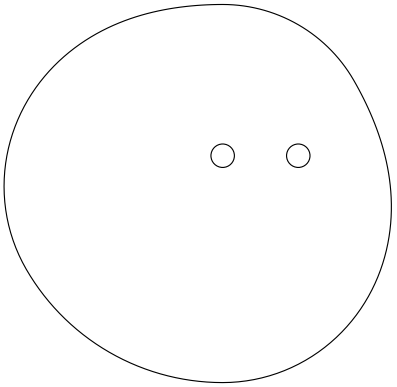
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

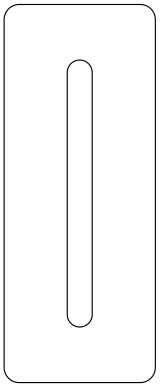
- Gap 8 mm. 4x M3 threaded rod 20 mm, 16x M3 nut, 16x M3 washer
- 3 mm smooth rod: 1x 24 mm (cam axle + handle), 2x 12 mm (follower guides),
- 1x 6 mm (cam lamination pin), 1x 15 mm (handle grip)



17-cam-plate x2



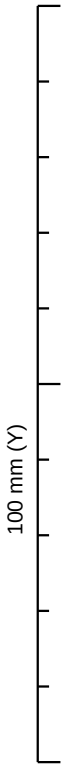
17-cam x3

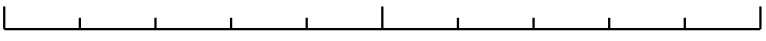


17-follower x3



17-handle



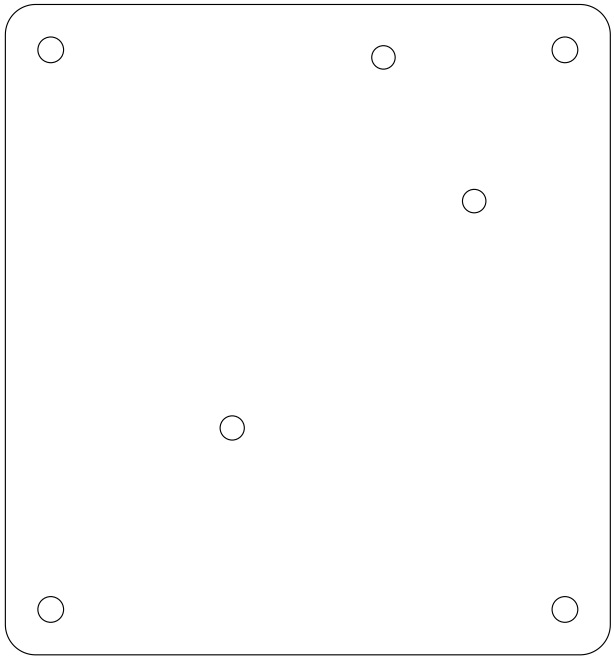


18 Ratchet between plates

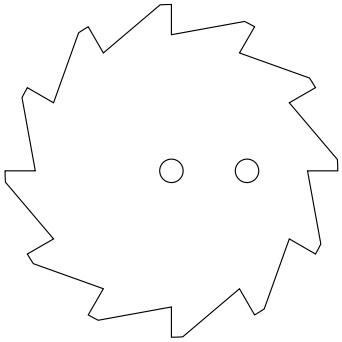
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

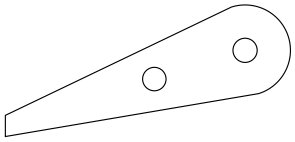
- Gap 8 mm. 4x M3 threaded rod 20 mm, 16x M3 nut, 16x M3 washer
- 3 mm smooth rod: 1x 24 mm (wheel axle + handle), 2x 12 mm (pawl pivot, spring
- anchor), 2x 6 mm (pins); small extension spring or rubber band



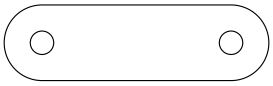
18-ratchet-plate x2



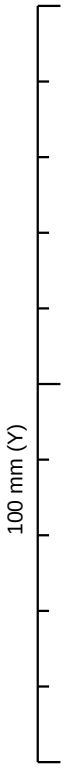
18-ratchet-wheel x3

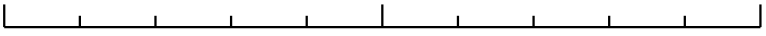


18-ratchet-pawl x3



18-handle



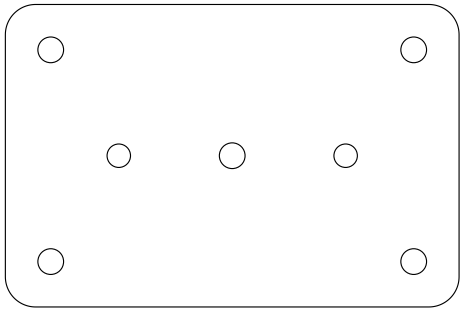


19 Lead-screw slide (M3 threaded rod drive)

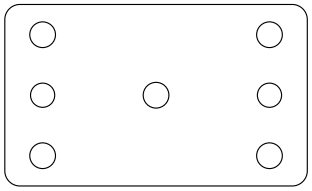
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

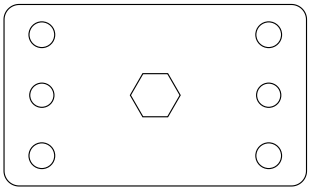
- Gap 80 mm. 4x M3 threaded rod 92 mm, 16x M3 nut, 16x M3 washer
- 1x M3 threaded rod 110 mm (lead screw) + 5x M3 nut, 2x 3 mm smooth rod 84 mm,
- 4x M3 x 10 screw + nut (carriage)



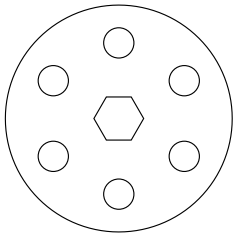
19-slide-end-plate x2



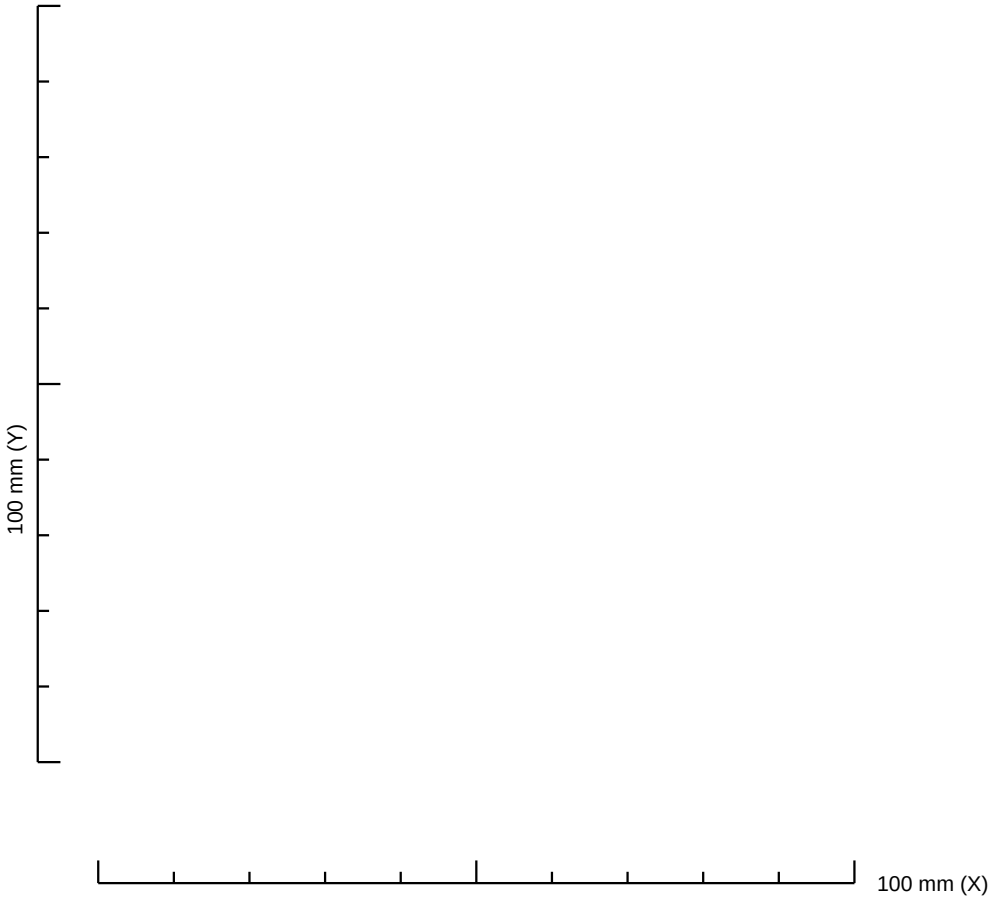
19-carriage-outer x2



19-carriage-nut



19-knob

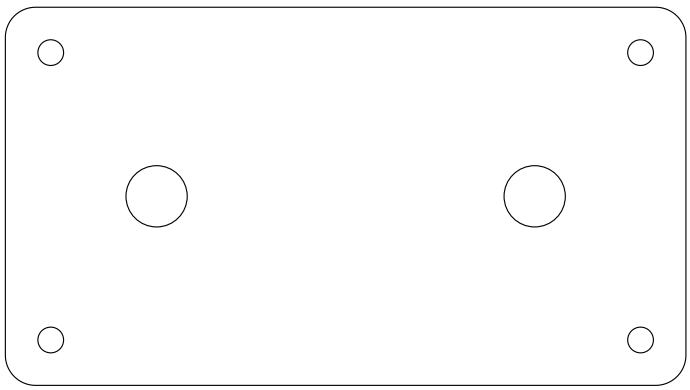


20 Linear rail: 8 mm rods + LM8UU carriage

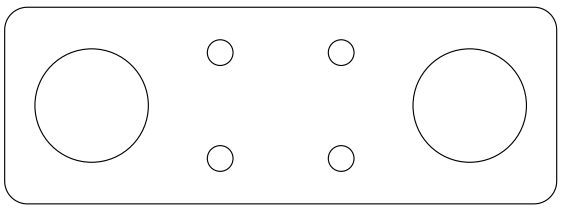
PRINT AT 100 % / ACTUAL SIZE (no 'fit to page'). Both bars must measure exactly 100 mm.
Black = cut, blue dashed = bend line, green = engraving. Material: 2 mm mild steel.

Hardware:

- Gap 150 mm. 4x M3 threaded rod 162 mm, 16x M3 nut, 16x M3 washer
- 2x 8 mm smooth rod 154 mm, 2x LM8UU, 4x M3 threaded rod 36 mm + 16 nuts (carriage)



20-rail-end-plate x2



20-carriage-plate x2

